Power Series and Taylor Series

ANSWER KEY

Radius and interval of convergence

1 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n}$.
Ratio test: $\left|\frac{x^{n+1}/(n+1)}{x^n/n}\right| \to |x|$. Converges for $|x| < 1$: radius of convergence $R = 1$.

2 Find the radius of convergence of $\sum_{n=0}^{\infty} \frac{x^n}{n!}$.
Ratio test: $\left|\frac{x^{n+1}n!}{(n+1)!\,x^n}\right| = \frac{|x|}{n+1} \to 0$ for every x: radius of convergence $R = \infty$.

Taylor and Maclaurin series

3 Find the first three nonzero terms of the Maclaurin series for $f(x) = e^x$, using $f^{(n)}(0) = 1$ for all n.
$e^x = 1 + x + \frac{x^2}{2!} + \cdots = 1 + x + \frac{x^2}{2} + \cdots$.

4 Find the third-degree Taylor polynomial for $f(x) = \cos x$ centered at $x = 0$.
$f(0) = 1$, $f'(0) = 0$, $f''(0) = -1$, $f'''(0) = 0$: $T_3(x) = 1 - \frac{x^2}{2}$.

5 Use the Maclaurin series $\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \cdots$ to approximate $\sin(0.2)$ using the first two nonzero terms, to four decimal places.
$0.2 - \frac{(0.2)^3}{6} = 0.2 - 0.001333\ldots \approx 0.1987$.

Interval of convergence (including endpoints)

6 Find the interval of convergence of $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n}$, checking both endpoints.
Ratio test gives radius $R = 1$, centered at $x = 2$: $1 < x < 3$ tentatively. At $x = 3$: $\sum \frac{1}{n}$ diverges (harmonic). At $x = 1$: $\sum \frac{(-1)^n}{n}$ converges (alternating). Interval of convergence: $[1, 3)$.

7 Find the radius of convergence of $\sum^{\infty} n!\, x^n$.
Ratio test: $\left|\frac{(n+1)!\,x^{n+1}}{n!\,x^n}\right| = (n+1)|x| \to \infty$ for any $x \neq 0$. The series converges only at $x = 0$: radius of convergence $R = 0$.

Using known series

8 Using the known Maclaurin series $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$, write the Maclaurin series for e^{-x^2} (substitute $-x^2$ for x), giving the first three terms.
$e^{-x^2} = 1 + (-x^2) + \frac{(-x^2)^2}{2!} + \cdots = 1 - x^2 + \frac{x^4}{2} - \cdots$.

9 Using $\frac{1}{1-x} = \sum^{\infty} x^n$ for $|x| < 1$, write the power series for $\frac{1}{1+x^2}$ (substitute $-x^2$ for x).
$\frac{1}{1+x^2} = \frac{1}{1-(-x^2)} = \sum_{n=0}^{\infty} (-x^2)^n = \sum_{n=0}^{\infty} (-1)^n x^{2n} = 1 - x^2 + x^4 - x^6 + \cdots$.

Taylor series centered at a nonzero point

10 Find the first three terms of the Taylor series for $f(x) = \ln x$ centered at $x = 1$, using $f(1) = 0$, $f'(1) = 1$, $f''(1) = -1$.

$T(x) = f(1) + f'(1)(x-1) + \frac{f''(1)}{2!}(x-1)^2 + \cdots = (x-1) - \frac{(x-1)^2}{2} + \cdots$.

11 Find the second-degree Taylor polynomial for $f(x) = \sqrt{x}$ centered at $x = 4$, given $f(4) = 2$, $f'(4) = \frac{1}{4}$, $f''(4) = -\frac{1}{32}$.

$T_2(x) = 2 + \frac{1}{4}(x-4) - \frac{1}{64}(x-4)^2$.

Error estimation with Taylor polynomials

12 Using the second-degree Maclaurin polynomial for $\cos x$, $T_2(x) = 1 - \frac{x^2}{2}$, estimate $\cos(0.3)$ to four decimal places, and compare to the true value $\cos(0.3) \approx 0.9553$.

$T_2(0.3) = 1 - \frac{0.09}{2} = 1 - 0.045 = 0.9550$, very close to the true value 0.9553 (error ≈ 0.0003).

Visualizing a Taylor approximation

13 The graph compares $f(x) = \cos x$ to its second-degree Maclaurin polynomial $T_2(x) = 1 - \frac{x^2}{2}$. Use the graph to describe where the approximation is best and where it breaks down.

The two curves are nearly indistinguishable near $x = 0$, where the approximation is best (the region of the expansion point). As $|x|$ increases toward $\pm\pi$, $T_2(x)$ (a downward parabola) diverges sharply from the bounded, oscillating $\cos x$, showing the approximation breaks down away from the center.

Differentiating and integrating power series

14 Given $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ for $|x| < 1$, differentiate both sides term-by-term to find a power series for $\frac{1}{(1-x)^2}$.

$\frac{d}{dx}\left[\frac{1}{1-x}\right] = \frac{1}{(1-x)^2}$. Differentiating the series: $\sum_{n=1}^{\infty} nx^{n-1}$. So $\frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} nx^{n-1} = 1 + 2x + 3x^2 + \cdots$.

15 Using the Maclaurin series $\sin x = \sum \frac{(-1)^n x^{2n+1}}{(2n+1)!}$, integrate term-by-term to find the first three terms of the power series for $\int \sin x \, dx = -\cos x + C$, and confirm they match the known Maclaurin series for $-\cos x$.

$\int \left(x - \frac{x^3}{6} + \cdots\right)dx = \frac{x^2}{2} - \frac{x^4}{24} + \cdots + C$. Comparing to $-\cos x = -1 + \frac{x^2}{2} - \frac{x^4}{24} + \cdots$: matching term-by-term (aside from the constant -1, absorbed into C) confirms consistency.

Using Taylor series to evaluate a limit

16 Use the Maclaurin series $\sin x = x - \frac{x^3}{6} + \cdots$ to evaluate $\lim_{x\to 0}\frac{\sin x - x}{x^3}$.

$\sin x - x = -\frac{x^3}{6} + \frac{x^5}{120} - \cdots$, so $\frac{\sin x - x}{x^3} = -\frac{1}{6} + \frac{x^2}{120} - \cdots \to -\frac{1}{6}$ as $x \to 0$.

17 Find the fourth-degree Taylor polynomial for $f(x) = e^x$ centered at $x = 0$, and use it to estimate $e^{0.5}$, comparing to the true value $e^{0.5} \approx 1.6487$.

$T_4(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24}$. $T_4(0.5) = 1 + 0.5 + 0.125 + 0.02083 + 0.0026 \approx 1.6484$, very close to the true value 1.6487.

18 An engineer needs a quick approximation for $\ln(1.1)$ without a calculator. Using the Maclaurin series $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$ for $|x| < 1$, find the approximation using the first three terms with $x = 0.1$, and compare to the true value $\ln(1.1) \approx 0.09531$.

$\ln(1.1) \approx 0.1 - \frac{(0.1)^2}{2} + \frac{(0.1)^3}{3} = 0.1 - 0.005 + 0.000333 \approx 0.095333$, extremely close to the true value 0.09531 (error ≈ 0.00002).